



● SENTINEL

The NETRA Platform · Unified CCTV Intelligence for Law Enforcement

An integrated video-management & analytics platform that onboards diverse CCTV systems, correlates live feeds with government databases, and turns them into real-time, auditable intelligence.

THE CHALLENGE

Thousands of cameras. No single brain.

Gujarat's cameras are spread across departments, vendors and protocols — each an island. There is no unified way to view them, correlate a vehicle across them, or act in real time.



Fragmented

IP, analog, and multi-vendor VMS systems that don't talk to each other



Blind between cameras

A vehicle seen at one junction can't be followed to the next



Video-heavy & unscalable

Centralising raw video for a whole state is physically infeasible

~80,000

cameras to integrate across Gujarat

The scale is the whole problem — the solution has to be built for it from the first camera, not retrofitted.

OUR APPROACH

Models 1, 2 & 3 — as one layered platform

In the field these aren't alternatives — they're layers of the same system. We built them as one.

MODEL 1

System of record

Camera registry, GIS map, jurisdiction hierarchy, RBAC, watchlist & alerts, and a tamper-evident audit trail — the mandatory spine both other layers depend on.

MODEL 3

Integration substrate

An adapter framework that onboards heterogeneous cameras and VMSs (ONVIF/RTSP, Genetec, Milestone, the government grid) behind one interface.

MODEL 2

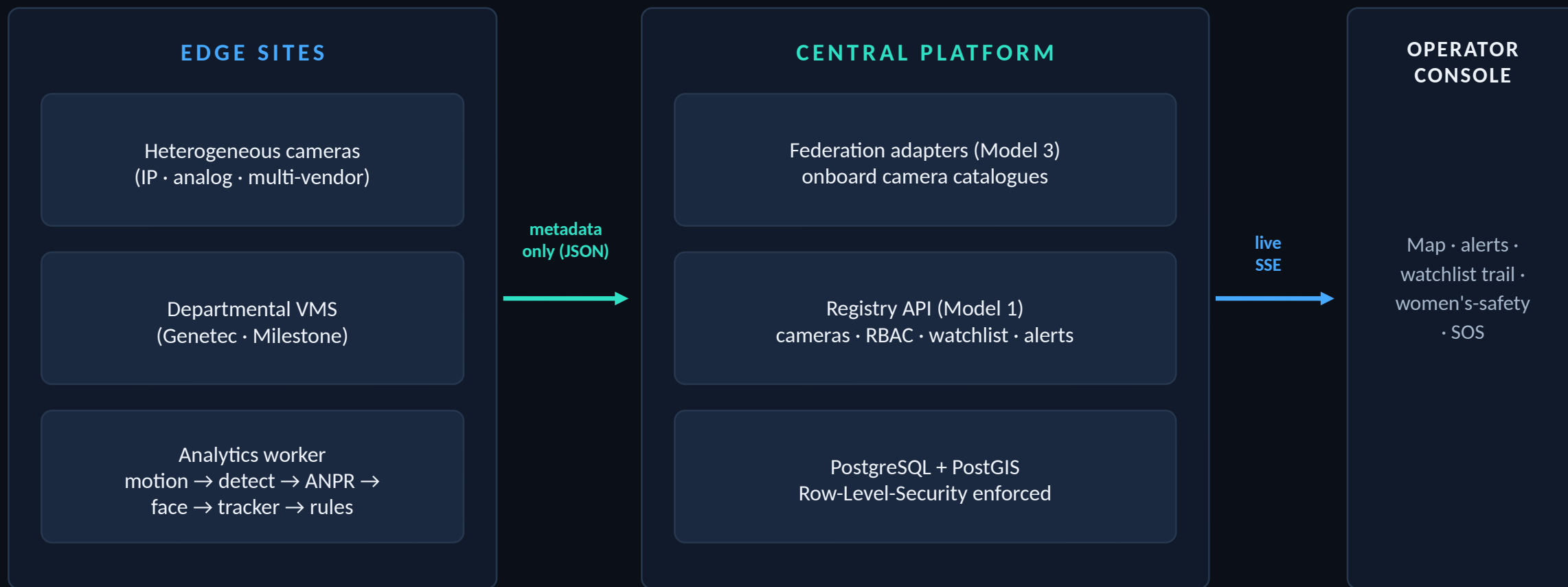
Edge analytics

Unified viewing metadata, ANPR, watchlist matching and alerts — inference runs at the edge, only metadata travels centrally.

Why one platform: you can't correlate a plate across cameras without one trusted registry of which camera is where, in whose jurisdiction, and who may see it.

ARCHITECTURE

Analyse at the edge, move only metadata



Video-central would be ~200 Gbps for 80,000 cameras — impossible. Metadata-central is a few Mbps.

One interface for every camera and VMS

Real deployments are never one vendor. Each source is normalised into one camera record, so everything downstream is protocol-agnostic.

● ONVIF / RTSP

IP cameras & NVRs, and analog behind an encoder — the direct-connect (Model 2) path

● Genetec

Departmental VMS integrated via its API — for cameras a dept won't expose directly

● Milestone

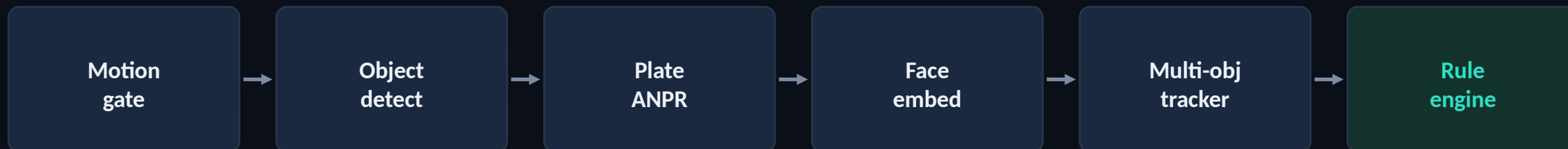
The other common VMS platform, behind the same driver interface

● Sentinel Grid

The government camera grid, consumed from its cameras.json catalogue

Catalogue-driven onboarding: one command reads a department's camera catalogue and registers the whole set — with locations and stream endpoints — in a single idempotent run.

A cost-ordered cascade — cheap rejection first



MEASURED

~8

OCR calls per 1,000 frames

A geometric plate gate + once-per-track dedupe spend the expensive stages on almost nothing — 1,920 detected objects → 4 OCR calls in the self-test.

ANPR built for evidence

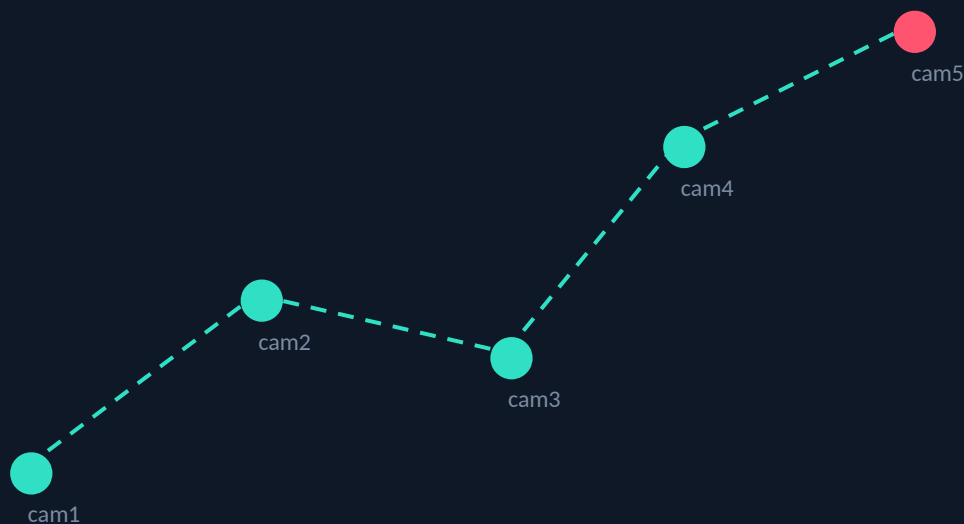
- **No character substitution** — OCR confusions are never 'fixed' — a substituted character is a fabricated one
- **Raw + normalised kept** — both forms stored on every read, so the evidence is auditable
- **Read once per vehicle** — not once per frame — no contradictory plate strings

THE LIVE TEST CASE

Track a vehicle across the whole grid

Given a plate on the day, Sentinel returns its complete, timestamped, location-wise route across cameras — the graded challenge, already working.

Same plate, two+ cameras, two+ times = a trail



How — with no new subsystem

Each watchlist match records camera + time. Two matches for one plate are, by construction, two points on the map — a trail. The console computes speed & heading between them. No biometric identity resolution.

VERIFIED LIVE

Two sightings of one stolen-plate entry from two cameras collapsed into one alert with a two-point trail, and the console rendered the route with estimated speed and heading — tested against a real database.

A dedicated women's-safety response path



Instant SOS — no login

POST /api/sos opens a critical alert immediately. A kiosk or a phone in a crisis can't be expected to authenticate first.



Behavioural, not appearance

A lone person followed, or encircled, in a flagged zone or after dark — geometry and time, never demographics.



Safety Mode on the console

Reprioritises SOS & safety alerts and highlights isolated/high-risk zones on the map at a glance.

Additive to the same alert pipeline — an SOS gets the audit trail and live push for free, not a bolted-on parallel system.

Authorization enforced by the database

A bug or compromise in the API cannot widen access — PostgreSQL decides, not the application.



One claim, derived authority

The app asserts only who the user is; jurisdiction, permissions and scope are derived in-database by SECURITY DEFINER functions. No broader claim exists to forge.



Tamper-evident audit trail

Append-only, hash-chained by database trigger, independently re-verifiable. Records purpose + case reference per access — DPDP Act 2023 aligned.



Credentials encrypted at rest

Per-credential keys wrapped by a master key held only in the environment; a DB-only compromise yields no usable passwords.

198

adversarial security tests

Forged tokens, credential transplant, audit tampering, timing oracles — the attacks, not just round-trips.

BUILT FOR THE NUMBER

Scaling to 80,000 cameras

VIDEO-CENTRAL

~200 Gbps

continuous ingress for 80,000 streams — physically infeasible

METADATA-CENTRAL (OURS)

a few Mbps

only alerts & matches travel centrally — ~16 GB/day statewide

Edge compute

~25–50 cameras / edge box → ~1,600–3,200 boxes, per district, linear

Flat security cost

RLS filters on indexed columns — query plans don't change from 5k to 80k

Storage & rollout

metadata-only, monthly-partitioned; phased on the jurisdiction tree

NOT SLIDWARE — RUNNING SOFTWARE

Built & verified

240

automated tests passing

198 security + 42 analytics

30

government cameras onboarded

across 10 Gujarat districts, live

3

reference models

delivered as one platform

0

video leaves the edge

metadata-only, by design



● See it live

Sign in → onboard the government grid → watch a plate light up across cameras → pull its route with speed & heading → raise an SOS.
